

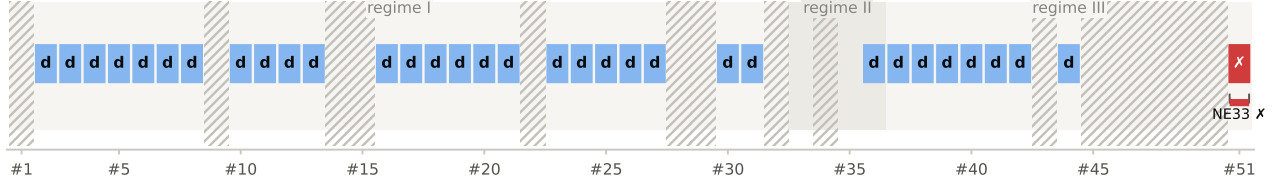
H81: Culture beyond composition: the emergent slow mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. In regime I the village content has a collective slow mode ($\tau_u \approx 25$ days) beyond its members and the measured goal fields, carried across turnover; it is absent inside #51 and does not jump at roster events.

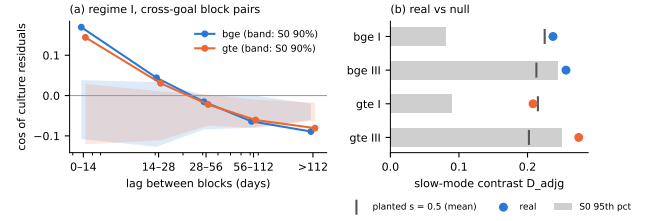
level: descriptive · pipeline checks 100% · updated 2026-10-04



Goal periods tested (2 failed, 32 descriptive). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The test asks whether the village content vector carries a slow component that its members do not explain. The composition null is each member's personal vector, fitted on its other goal periods, plus the measured goal fields (kickoff, goal text, room kickoffs, human messages). The residual is compared between blocks of different goal periods. Regime I (24 goals, May 2025 to February 2026) shows a slow mode. Blocks at most 14 days apart align ($\bar{s} = 0.17$, bge). The alignment decays with $\tau_u \approx 23$ –28 days. The contrast is 3 times the synthetic null in both embedding models. It survives member turnover, dropping Gemini 2.5 Pro, and the scaffold steps. Regime III is marginal. Inside #51 the common mode lives about 3 days. Roster joins do not move it. The holdout script exists and is not run.



Observables. (a) Regime I: mean similarity of culture residuals between blocks of different goal periods against their lag. Bands are the 90% range of the synthetic composition null (S0) on the real panel. Similarity is high below two weeks and decays over about a month. (b) The slow-mode contrast D_{adjg} per model and regime (dots), the S0 95th percentile (grey bars) and the planted mean for a slow mode with half the period-field variance (ticks).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect a village to drift as a whole on a timescale of about a month, independent of who is in it. In regime I, half of the period-to-period content variance beyond the goal fields is this slow mode. An operator who wants a clean goal change should not expect the new kickoff to reset the village: neighbouring goals share about 0.17 of their residual direction. The mode does not respond to roster changes, so adding or retiring agents is not a lever on it. Its carrier is unknown. It may be the shared record (sites, documents, repos) or an outside drift that every agent reads. For monitoring, compare a village with its own last two to three weeks, not with its whole history.

PHYSICAL MODEL

Vector spins (model 11). The spin $v_{i,d}$ is agent i 's style-residualized content vector on day d (32-d). a_i is the personal vector (weights, family, style). h_G is the exogenous field of goal period G , removed by projection Π_G . w_b is a topic field with no memory. u_b is the collective slow mode, an Ornstein–Uhlenbeck vector with correlation time τ_u . The order parameter is the block culture residual u_b . The statistic D is the near-minus-far similarity of u_b across goal boundaries, adjusted for goal similarity (D_{adjg}) and member overlap (D_{adj}).

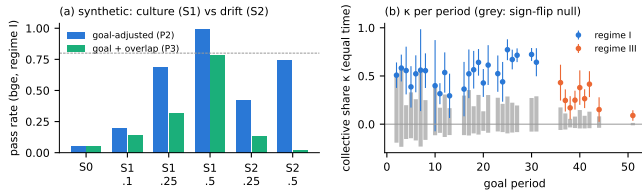
$$v_{i,d} = a_i + h_{G(d)} + w_{b(d)} + u_{b(d)} + e_{i,d}, \quad \langle u_b \cdot u_{b'} \rangle \propto e^{-|t_b - t_{b'}|/\tau_u}$$

$$r_{i,b} = \Pi_G(\bar{v}_{i,b} - \hat{a}_i^{(-G)}), \quad u_b = \langle r_{i,b} \rangle_{i \in b}$$

$$D = \langle \cos(u_b, u_{b'}) \rangle_{\Delta t \leq 14 \text{ d}} - \langle \cos(u_b, u_{b'}) \rangle_{\Delta t \geq 42 \text{ d}}$$

RESULTS VS PREDICTIONS

Prediction	Observed (bge / gte)	Verdict
P1 equal-time share $\kappa > \text{null}$	29/33, 31/33 periods; median 0.51 / 0.53 (field-dominated)	pass
P2 slow mode $D_{\text{adj}} > S0$ q95	I: 0.237 / 0.208 (q95 0.081 / 0.089); III: 0.256 / 0.275 (q95 0.245 / 0.250)	pass (I)
P2 decay over weeks	$\tau_u = 28 \pm 5$ / 23 ± 4 d	pass
P3 across turnover (D_{adj})	I: 0.166 / 0.170 (q95 0.10); size-matched disjoint 0.188 / 0.156	pass (I)
P4 jumps at roster events	percentile 0.08 / 0.14 over 22 events	fail
P5 without Gemini 2.5 Pro	$\times 1.06$ / $\times 1.06$; size-matched $\times 0.90$ / $\times 0.96$	pass
N1 G51 slow mode inside #51	$L_{\text{slow}} = -0.005$; common mode $\tau \approx 3$ d	fail
N2 NE33/NE32 batch joins	stayer-jump percentiles 0.14–0.89	fail
PH scaffold steps, 3 fields	within segments 0.157 / 0.154; post hoc 3 covariates 0.237 / 0.209	



Second figure. (a) Synthetic pass rates on the real regime-I panel. The goal-adjusted contrast detects a slow mode (S1) and also member drift (S2). The overlap-adjusted contrast rejects drift. (b) Equal-time collective share κ per goal period with the sign-flip null (grey).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive

+24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only measured exogenous directions are removed. A slow outside drive that every agent reads would look the same. The HH’s literal composition null is biased: it produced +0.37 in regime-III synthetics, so the estimator uses two-way fixed effects (Amendment A1). Power is 0.96–0.99 for a slow mode with half the period-field variance and 0.67–0.69 at a quarter. Regime III has 9 goals and power ≤ 0.42 . Regime I has 4–12 agents. The turnover-disjoint statistic has low power unless size-matched. τ_u is a descriptive fit.

ROUND 2

Name the carrier: regress the slow mode on long-lived artifacts that agents read (context ledger) against posted-but-unread artifacts. Probe outside drift with dated external topics. Fit one OU model to the H81 profile and H82’s boundary trace.

- **Essence.** a village-level state variable that persists for weeks and survives turnover, and what carries it.
- **H81-R1** Name the carrier.
- **H81-R2** Exogenous drift probe.
- **H81-R3** Regime III with more units.
- **H81-R4** Link to H82.